

Testnet Infrastructure Validation Report

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Environment	Dual Node — Windows (RPC :8545) + WSL Ubuntu (RPC :8546) — Indonesia / CGNAT
Report Type	Infrastructure Validation — RPC Stability & Transaction Reliability

Executive Summary

This report documents follow-up infrastructure validation testing conducted on DAC Testnet, focusing on RPC stability, wallet interoperability, EIP-1559 compatibility behavior, and transaction propagation reliability. Testing was performed using a dual-node single-machine environment running simultaneously on Windows (port 8545) and WSL Ubuntu (port 8546), operating under a CGNAT residential network in Indonesia.

Finding	Wallet Tested	Result
Localhost RPC outperforms public RPC	All wallets	Confirmed
Consistent transaction success via localhost RPC	OKX Wallet	Confirmed
Transactions frequently pending — EIP-1559 type-2 behavior	MetaMask	Pending
Improved compatibility and transaction success	Rabby Wallet	Confirmed
eth_feeHistory warnings — EIP-1559 incomplete implementation	Node logs	Error observed

Environment Overview

Testing was conducted using a dual-node architecture running simultaneously on a single machine under CGNAT residential network constraints in Indonesia. Each node exposed a local HTTP RPC endpoint and was connected to DAC Testnet authority peers.

Node	Platform	RPC Endpoint	Wallet Tested
Windows Node	Windows Host	localhost:8545	OKX Wallet
WSL Node	WSL Ubuntu	localhost:8546	MetaMask / Rabby

Key Technical Findings

RPC Stability — Localhost vs Public Endpoint

Self-hosted localhost RPC endpoints performed significantly better than the official public RPC endpoint (rpctest.dachain.tech). The public endpoint continued to exhibit gateway timeout behavior, consistent with findings documented in the previous report (05 May 2026).

RPC Endpoint	Stability	Transaction Result
localhost:8545 (Windows node)	Stable	Consistent success (OKX)
localhost:8546 (WSL node)	Stable	Rabby: success / MetaMask: pending
rpctest.dachain.tech (public)	Unstable	Gateway timeout — 504

Wallet Interoperability

Clear performance differences were observed across wallet clients under identical node and network conditions. The divergence is reproducible across multiple test iterations and is attributable to transaction format differences, not infrastructure issues.

Wallet	Result	Notes
OKX Wallet	Confirmed	Consistent success via localhost RPC — legacy tx format
Rabby Wallet	Confirmed	Significantly improved compatibility over MetaMask
MetaMask	Pending	EIP-1559 Type-2 tx — repeated eth_feeHistory warnings

Technical Observation: EIP-1559 Compatibility

MetaMask submitted Type-2 (EIP-1559) transactions using maxFeePerGas and maxPriorityFeePerGas fields. Repeated warnings observed from DAC node logs suggest possible incompatibility or incomplete implementation of eth_feeHistory handling.

```
err="invalid argument 0: json: cannot unmarshal string into Go value of type int"
```

This error is generated every time MetaMask queries eth_feeHistory before submitting a transaction. The node handler expects an integer type but receives a string — indicating a type mismatch in the EIP-1559 fee history implementation. As a result, MetaMask's fee estimation operates on degraded or fallback data, producing transaction parameters that DAC validators do not reliably include in blocks.

Operational Evidence

- Repeated successful transaction submissions were recorded via localhost RPC endpoints on both nodes.
- Chain segments were successfully imported after transaction propagation, confirming end-to-end delivery.

- Wallet interoperability differences were reproducible across multiple test iterations.
- Chain reorg events were also observed during network activity — consistent with normal testnet behavior.

Selected Transaction Submission Logs

```
INFO Submitted transaction
hash=0x1c6e99e702870f9ab59ba67a2b721f29b3974e19a6053d2cb3ceb382f158d350

INFO Submitted transaction
hash=0x1ad4538337dc14232bd6bf87971fbb016e5b915b79556769ea9f03f00876951f

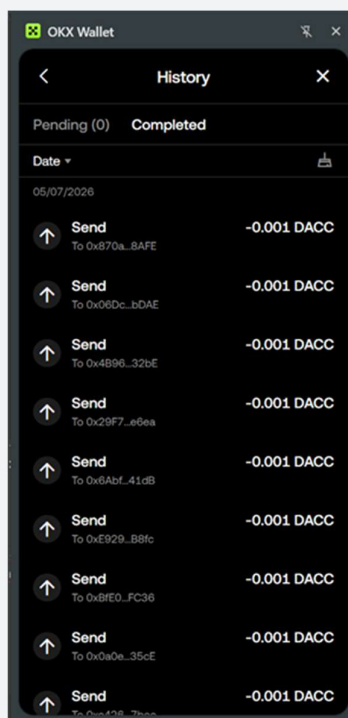
INFO Submitted transaction
hash=0x75bed5524d3fbe64a44a56f8d3ccb99cb39b3b2057b62dcee4a0846d404d6186

INFO Submitted transaction
hash=0x36fd2bbc821976aa49d3e74e203b0b8b85d1d9e96cee38fc7cd98714e2e84301
```

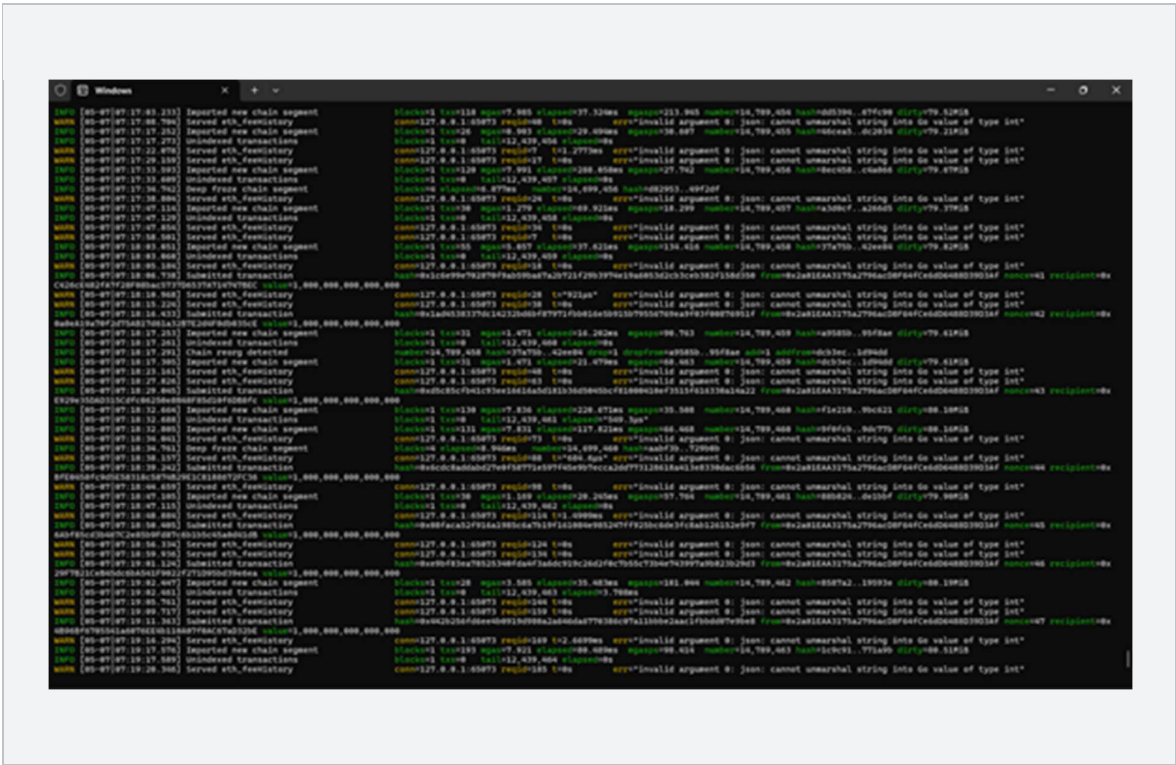
Supporting Screenshots / Proof

The following placeholders mark where supporting screenshots should be inserted. Each screenshot provides visual proof of the findings described in this report.

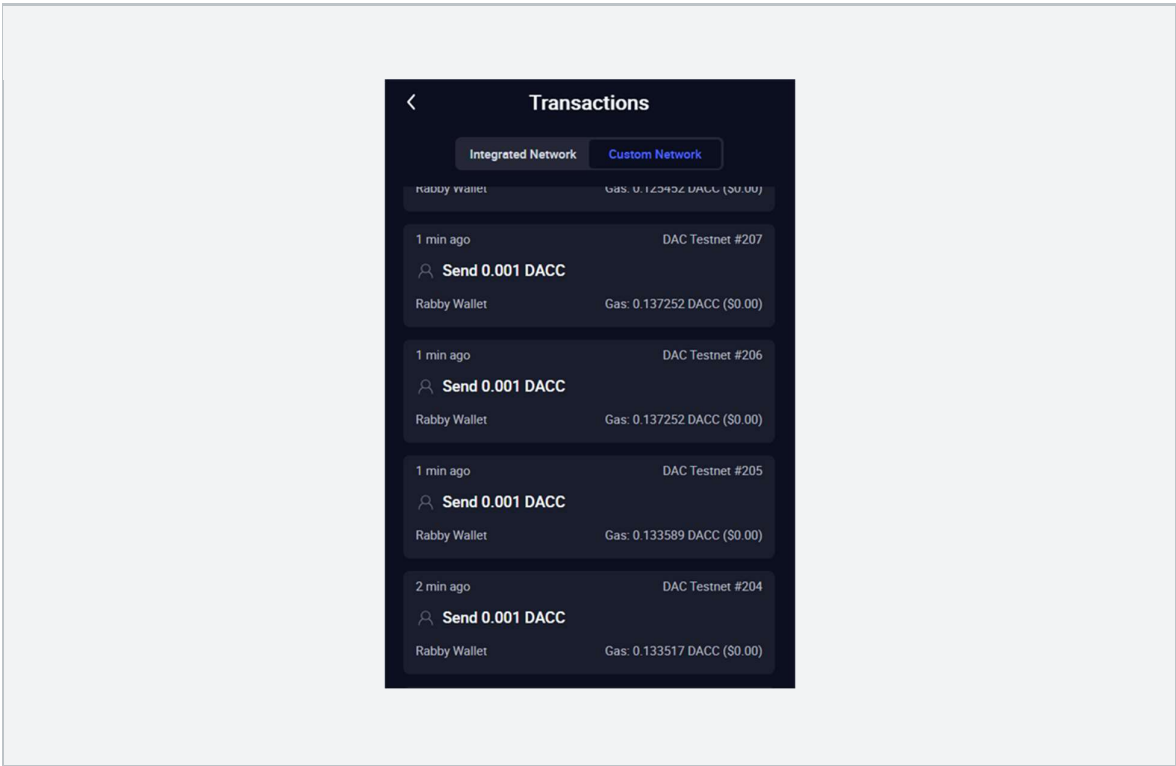
Screenshot 1 — OKX Wallet — Successful transactions via localhost RPC



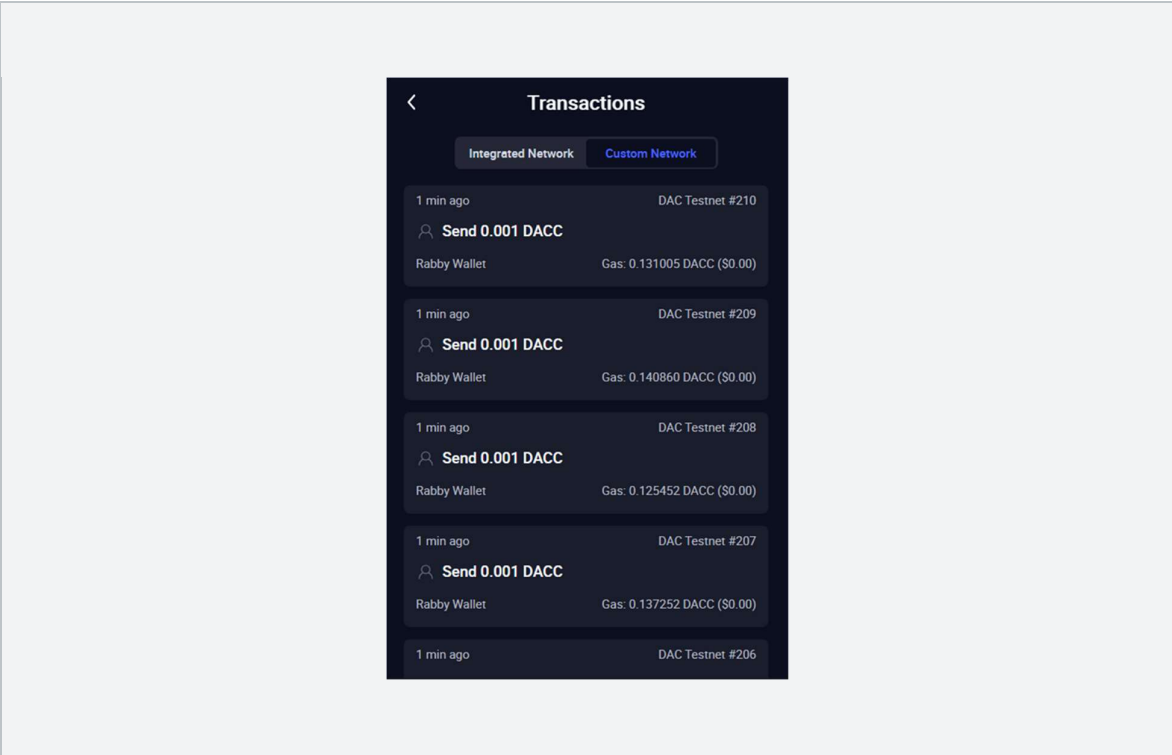
Screenshot 2 — Windows node — Operational logs and transaction submission activity



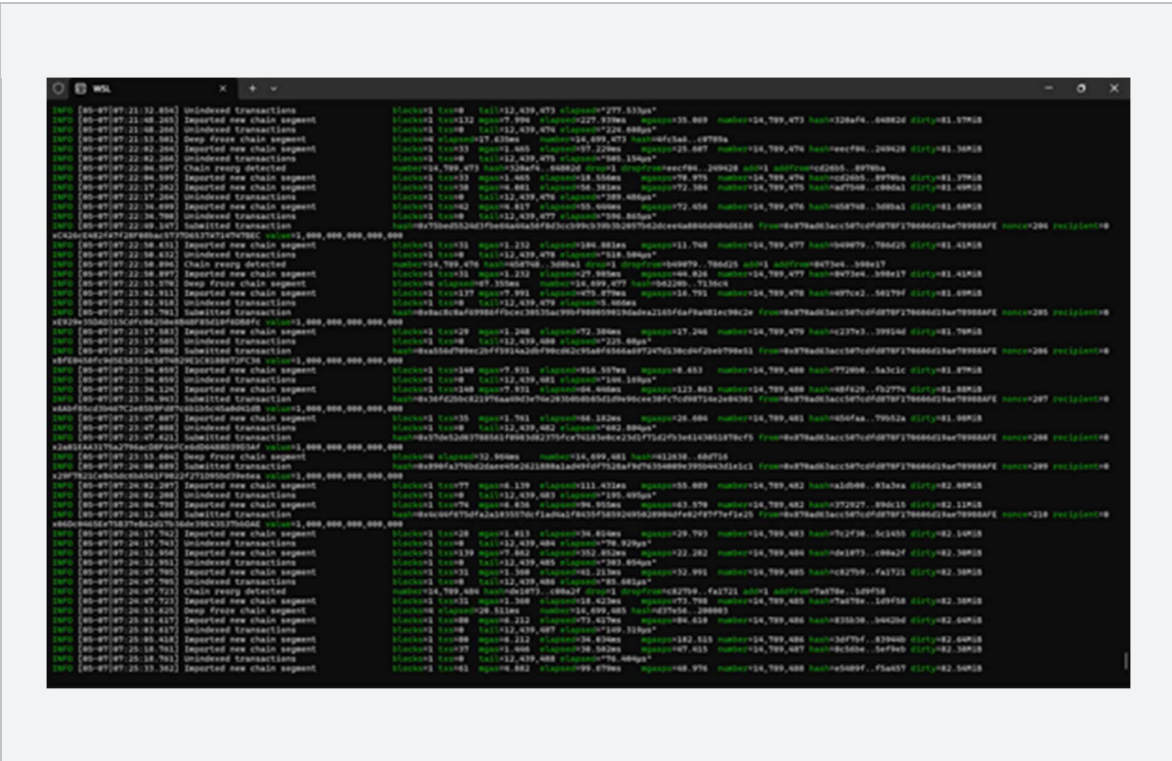
Screenshot 3 — Rabby Wallet — Successful transaction execution



Screenshot 4 — Rabby Wallet — Additional transaction execution proof



Screenshot 5 — WSL node — Operational logs and transaction propagation activity



Recommendations

For DAC Team

- Audit EIP-1559 compatibility across validators and RPC middleware.
- Review and fix eth_feeHistory implementation — resolve the unmarshal type mismatch.
- Validate txpool propagation handling for Type-2 (EIP-1559) transactions.
- Expand and stabilize public RPC backend infrastructure to reduce dependency on a single endpoint.
- Provide temporary wallet compatibility guidance for users and developers.
- Implement monitoring for RPC latency, timeout rate, and txpool consistency.

For Testnet Participants

- Use OKX Wallet or Rabby Wallet as primary wallets for DAC Testnet — avoid MetaMask until EIP-1559 compatibility is confirmed.
- Run a local node and connect wallets to a self-hosted localhost RPC for improved reliability.

Conclusion

Component	Status
DAC Testnet consensus / P2P layer	Operational — block production ongoing
Localhost RPC (both nodes)	Stable — confirmed via OKX and Rabby
Public RPC endpoint	Unstable — accessibility challenges persist
EIP-1559 transaction compatibility	Incomplete — MetaMask Type-2 tx not reliably included

Current findings strongly indicate that DAC Testnet consensus functionality remains operational, while the public RPC layer and EIP-1559 transaction compatibility behavior continue to present accessibility and transaction reliability challenges for end users.